

48 V to isolated 12 V / 5 A • Revision A1 engineering prototype • 2026-10-10

[illegible]

$C_r = C_7 + C_8 = 200 \text{ nF}$; $f_r \approx 193 \text{ kHz}$ including 0.1 uH leakage allowance

T1: two 0.05 mm gapped ELP22 N87 halves; four primary turns, two per half-secondary.

Primary series midpoint (pin 6) is internal winding copper; no external connection.

8 layers / 70 um copper requested. Functional isolation only; no certified safety barrier.

[illegible][illegible]

R4/R5 and R6/R7 translate 12 V GD outputs to 3.7 V logic. Do not bypass

12.05 V nominal feedback; 13.27 V nominal OVP. Compensation and light-load behavior require bench tuning

[illegible]

Planar Studio / KiStack workflow

Sheet: /
EN - DE CAN COMU - L - A

File: PS-GAN-60W.kicad_sch

Title: PS-GAN-60W: planar LLC engineering prototype

Size: A2	Date: 2026-10-10
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Rev: A1-review

REVIEW ONLY – NOT RELEASED FOR FABRICATION. See project review for sourcing, electrical assumptions and validation status.